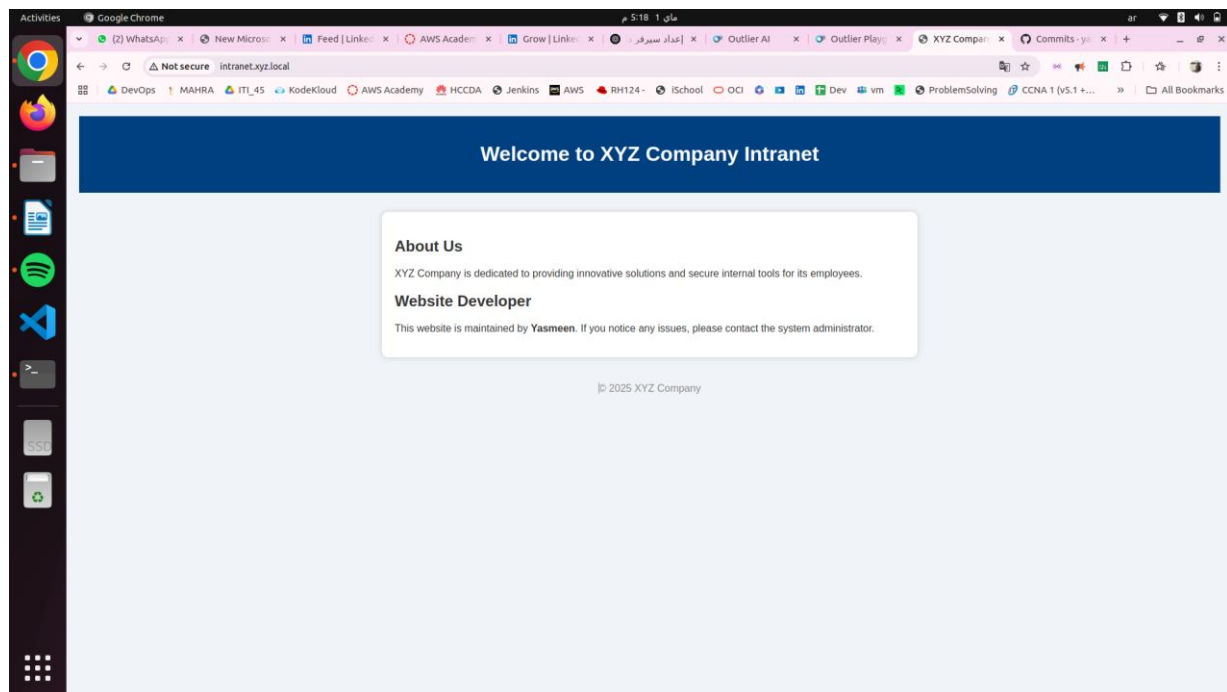


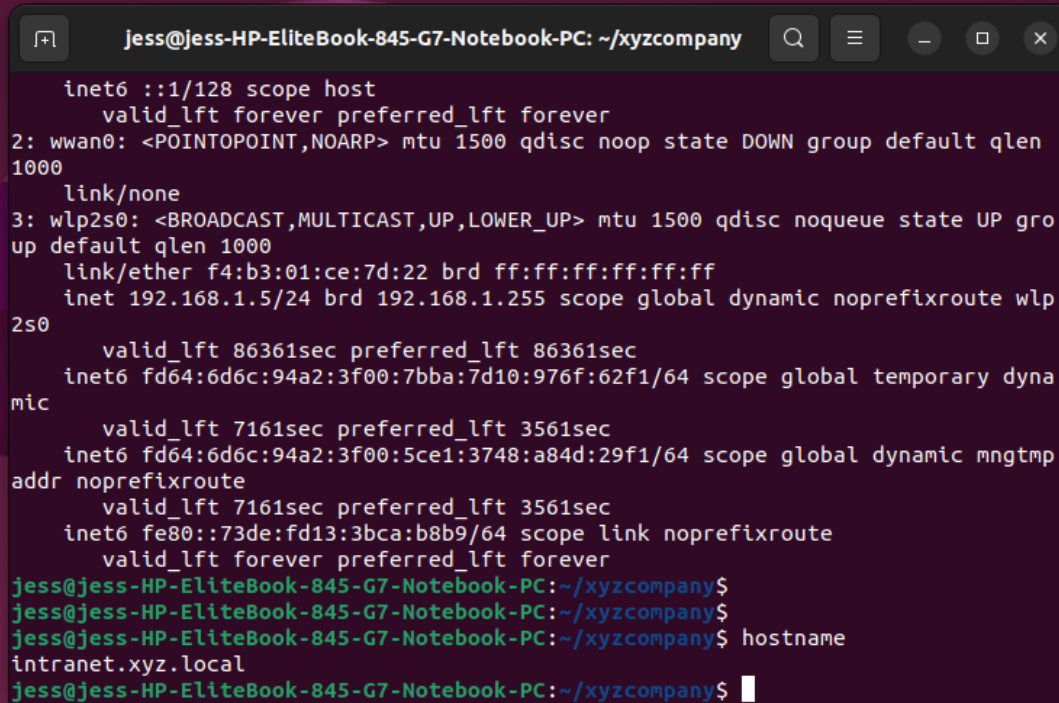
How Setting Up an Internal Web Server for XYZ Company ?

XYZ Company needs an **internal Web Server** to display company information and share some files among employees. As a **System Administrator**, your task is to set up the environment, run the website, and manage permissions securely.



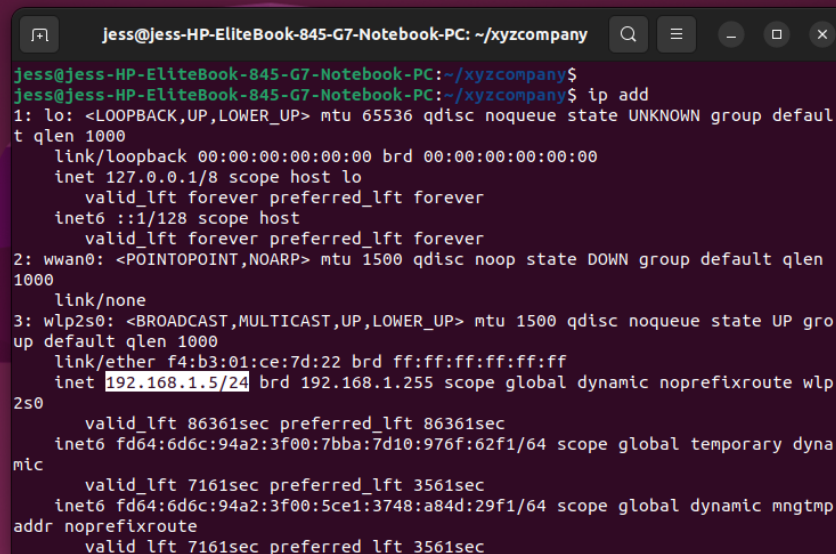
1. Configure Network and Connectivity:

- ✓ Set the **hostname** to intranet.xyz.local



```
jess@jess-HP-EliteBook-845-G7-Notebook-PC: ~/xyzcompany
inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: wwan0: <POINTOPOINT,NOARP> mtu 1500 qdisc noop state DOWN group default qlen
1000
    link/none
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP gro
up default qlen 1000
    link/ether f4:b3:01:ce:7d:22 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.5/24 brd 192.168.1.255 scope global dynamic noprefixroute wlp
2s0
        valid_lft 86361sec preferred_lft 86361sec
    inet6 fd64:6d6c:94a2:3f00:7bba:7d10:976f:62f1/64 scope global temporary dyna
mic
        valid_lft 7161sec preferred_lft 3561sec
    inet6 fd64:6d6c:94a2:3f00:5ce1:3748:a84d:29f1/64 scope global dynamic mngtmp
addr noprefixroute
        valid_lft 7161sec preferred_lft 3561sec
    inet6 fe80::73de:fd13:3bca:b8b9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ hostname
intranet.xyz.local
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
```

- ✓ Ensure the **IP Address** is **static** within the internal network



```
jess@jess-HP-EliteBook-845-G7-Notebook-PC: ~/xyzcompany
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: wwan0: <POINTOPOINT,NOARP> mtu 1500 qdisc noop state DOWN group default qlen
1000
    link/none
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP gro
up default qlen 1000
    link/ether f4:b3:01:ce:7d:22 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.5/24 brd 192.168.1.255 scope global dynamic noprefixroute wlp
2s0
        valid_lft 86361sec preferred_lft 86361sec
    inet6 fd64:6d6c:94a2:3f00:7bba:7d10:976f:62f1/64 scope global temporary dyna
mic
        valid_lft 7161sec preferred_lft 3561sec
    inet6 fd64:6d6c:94a2:3f00:5ce1:3748:a84d:29f1/64 scope global dynamic mngtmp
addr noprefixroute
        valid_lft 7161sec preferred_lft 3561sec
```

- ✓ Modify the `/etc/hosts` file to set up local **name resolution**.

```
jess@jess-HP-EliteBook-845-G7-Notebook-PC: ~/xyzcompany
valid_lft 7161sec preferred_lft 3561sec
inet6 fe80::73de:fd13:3bca:b8b9/64 scope link noprefixroute
valid_lft forever preferred_lft forever
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ hostname
intranet.xyz.local
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ Hostnamectl
Command 'Hostnamectl' not found, did you mean:
  command 'hostnamectl' from deb systemd (249.11-0ubuntu3.12)
Try: sudo apt install <deb name>
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ cat /etc/hosts
127.0.0.1        localhost
127.0.1.1        jess-HP-EliteBook-845-G7-Notebook-PC

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
192.168.1.5 intranet.xyz.local
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
```

2. Create Users and Assign Roles:

User	Role	Permissions
admin1	System Admin	Has sudo privileges to manage the server.
developer1	Web Developer	Can modify website files only. /var/www/html
viewer1	Regular Employee	Can only read shared files. /var/www/shared

- ✓ Create users using `useradd`. & Set passwords for each user.

```
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ cat /etc/passwd | grep '
admin1\\|developer1\\|viewer1'
admin1:x:1001:1001::/home/admin1:/bin/sh
developer1:x:1002:1002::/home/developer1:/bin/sh
viewer1:x:1003:1003::/home/viewer1:/bin/sh
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
```

- ✓ Add admin1 to the **sudo** group.

```
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ sudo -l -U admin1
[sudo] password for jess:
Matching Defaults entries for admin1 on intranet:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
    use_pty

User admin1 may run the following commands on intranet:
    (ALL : ALL) ALL
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
```

- ✓ Ensure developer1 can modify website files but has no administrative privileges.

```
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$ sudo -l -U developer1
User developer1 is not allowed to run sudo on intranet.
jess@jess-HP-EliteBook-845-G7-Notebook-PC:~/xyzcompany$
```

3. Set Up and Manage Files & Permissions:

- ✓ **Required File Structure:**

/var/www/html/	Main website files
/var/www/shared/	Shared company files

&

Required Permissions:

	Owner	Permissions
/var/www/html/	developer1	developer1 can modify files, while others can only read them
/var/www/shared/	admin1	All users can read, but cannot modify files

```
jess@hp-elitebook:~$ ls -ld /var/www/html
drwxr-xr-x 2 developer1 developer1 4096 17:00 1 ملى /var/www/html
jess@hp-elitebook:~$
```

```
jess@hp-elitebook:~$ ls -ld /var/www/shared
drwxr-xr-x 2 admin1 admin1 4096 16:06 1 ملى /var/www/shared
jess@hp-elitebook:~$
```

4. Install and Configure Web Server:

- ✓ Install Apache (httpd).

```
jess@hp-elitebook: ~  
jess@hp-elitebook:~$ ls -ld /var/www/html  
drwxr-xr-x 2 developer1 developer1 4096 17:00 1 ملى /var/www/html  
jess@hp-elitebook:~$ ls -ld /var/www/shared  
drwxr-xr-x 2 admin1 admin1 4096 16:06 1 ملى /var/www/shared  
jess@hp-elitebook:~$ systemctl status httpd  
Unit httpd.service could not be found.  
jess@hp-elitebook:~$ systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese>  
   Active: active (running) since Thu 2025-05-01 16:55:15 EEST; 3h 10min ago  
     Docs: https://httpd.apache.org/docs/2.4/  
  Process: 29707 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/S>  
 Main PID: 29711 (apache2)  
    Tasks: 55 (limit: 18283)  
   Memory: 6.0M  
      CPU: 519ms  
   CGroup: /system.slice/apache2.service  
           └─29711 /usr/sbin/apache2 -k start  
             └─29712 /usr/sbin/apache2 -k start  
               └─29713 /usr/sbin/apache2 -k start  
  
16:55:15 01 ملى intranet.xyz.local systemd[1]: Starting The Apache HTTP Server.>  
16:55:15 01 ملى intranet.xyz.local systemd[1]: Started The Apache HTTP Server.  
lines 1-16/16 (END)
```

✓ test HTML page in /var/www/html/index.html.

```

!DOCTYPE html
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>XYZ Company Intranet</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f4f8;
      color: #333;
      margin: 0;
      padding: 20px;
    }
    header {
      background-color: #004080;
      color: white;
      padding: 20px 0;
      text-align: center;
    }
    main {
      margin-top: 30px;
    }
    section {

```

"/var/www/html/index.html" [readonly] 61L, 1391B 1,1 Top